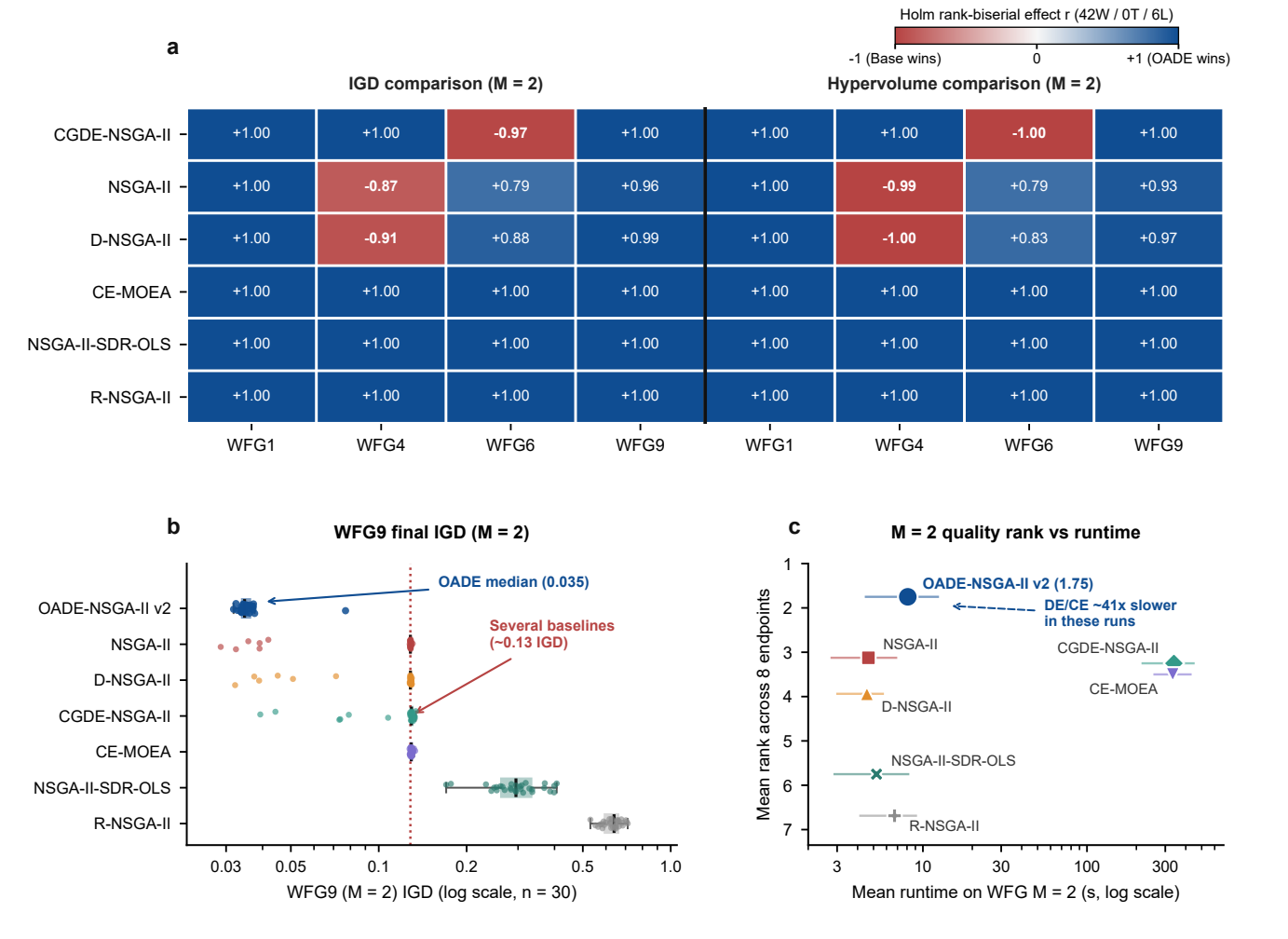

a
Holm rank-biserial effect r (42W / 0T / 6L)
-1 (Base wins)
0
+1 (OADE wins)
IGD comparison (M = 2)
Hypervolume comparison (M = 2)
CGDE-NSGA-II
NSGA-II
D-NSGA-II
CE-MOEA
NSGA-II-SDR-OLS
R-NSGA-II
+1.00 +1.00 -0.97 +1.00 +1.00 +1.00 -1.00 +1.00
+1.00 -0.87 +0.79 +0.96 +1.00 -0.99 +0.79 +0.93
+1.00 -0.91 +0.88 +0.99 +1.00 -1.00 +0.83 +0.97
+1.00 +1.00 +1.00 +1.00 +1.00 +1.00 +1.00 +1.00
+1.00 +1.00 +1.00 +1.00 +1.00 +1.00 +1.00 +1.00
+1.00 +1.00 +1.00 +1.00 +1.00 +1.00 +1.00 +1.00
WFG1 WFG4 WFG6 WFG9 WFG1 WFG4 WFG6 WFG9
b
WFG9 final IGD (M = 2)
OADE-NSGA-II v2
NSGA-II
D-NSGA-II
CGDE-NSGA-II
CE-MOEA
NSGA-II-SDR-OLS
R-NSGA-II
OADE median (0.035)
Several baselines
(~0.13 IGD)
0.03 0.05 0.1 0.2 0.5 1.0
WFG9 (M = 2) IGD (log scale, n = 30)
c
M = 2 quality rank vs runtime
Mean rank across 8 endpoints
1 2 3 4 5 6 7
OADE-NSGA-II v2 (1.75)
DE/CE ~41x slower
in these runs
NSGA-II
CGDE-NSGA-II
CE-MOEA
D-NSGA-II
NSGA-II-SDR-OLS
R-NSGA-II
3 10 30 100 300
Mean runtime on WFG M = 2 (s, log scale)